GradientCIO

REAL ASSETS PORTFOLIO CONSTRUCTION

Northstar Real Assets Portfolio

Portfolio construction review · prepared for the Investment Committee

AS OF
2026-06-30

TOTAL NAV
$1,000.0M

REAL-ASSETS WEIGHT
15.0%

UNFUNDED COMMITMENTS
$28.0M

01 Executive Decision

Valuation lag and structure mix need monitoring

RECOMMENDATION

We recommend retaining the 15.0% real-assets allocation while shifting 2.0 percentage points from listed real estate to infrastructure and natural resources across marketable and drawdown sleeves S3 S4. Committee action requested: approve the target structure and phased implementation, subject to valuation refresh, manager diligence and liquidity tests S3 S7 S8.

REAL-ASSETS WEIGHT	MARKETABLE SHARE	UNFUNDED COMMITMENTS	POLICY STATUS
15.0%	8.0%	$28.0M	Near limit
Policy target 15.0% S3	53.3% of real assets S4	2.8% of NAV S4	Private valuation lag S3 S4

Recommendation and action requested

We recommend a 2.0 percentage-point shift from listed real estate to infrastructure and natural resources while holding the 15.0% total allocation. Committee action requested: approve the target and staged implementation S3 S4 S8.

DECISION SUMMARY

DECISION ELEMENT	PROPOSED ACTION	EVIDENCE	STATUS
Total allocation	Hold at 15.0%	S3	Compliant
Listed real estate	Reduce from 5.0% to 3.0%	S3 S4	Proposed
Infrastructure and natural resources	Increase from 6.0% to 8.0%	S3 S4	Proposed

Conditions and limits

Refresh private valuations, confirm the commodity and inflation-linked mandate mappings, and complete manager diligence before funding S7 S8.

Key judgment — Valuation lag

Observation: Valuation lag and the listed-versus-private mix both need monitoring S1. Why it matters: The 2.0pp shift to infrastructure and natural resources depends on current private marks. Uncertainty: Private valuations are reported with a lag. What would change the view: Refreshed marks that change the relative weights.

Key judgment — Sleeve size

Observation: Real assets are 15.0% of total NAV S2. Why it matters: The proposal reshapes the sleeve without changing its total. Uncertainty: The weight moves with listed real estate prices. What would change the view: A drift of more than 1pp before funding.

Key judgment — Policy limit

Observation: The real-assets allocation is near its policy limit S3. Why it matters: Funding must be staged so the sleeve stays inside its range. Uncertainty: The limit test uses lagged private valuations. What would change the view: A valuation refresh that moves the sleeve away from the limit.

02 Mandate, Inflation Objective and Data Basis

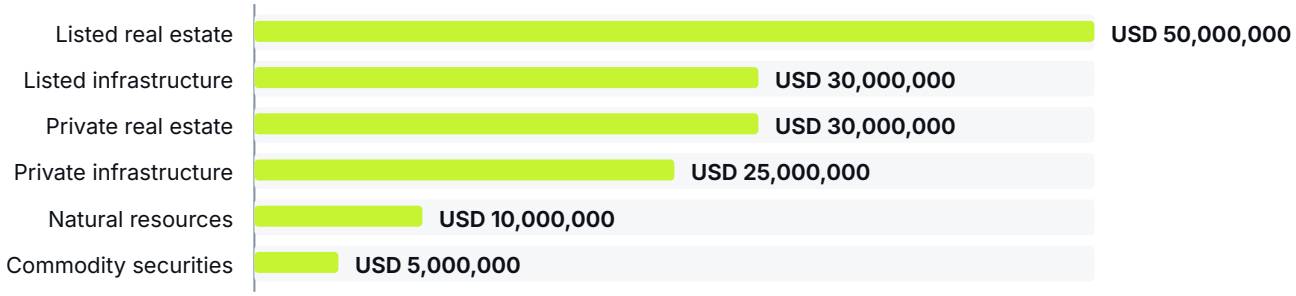
Portfolio	**Northstar Total Fund** S2
Mandate	**Broad real-assets diversification** S8
Inflation objective	**Participate in inflation-sensitive cash flows without a guaranteed outcome** S8
Horizon	**10 years** S6
Base currency	**USD** S2
Data scope	**Illustrative sample data** S1 S1

GOVERNED FRAMEWORK

AREA	OBJECTIVE OR LIMIT	OBSERVED	STATUS	SOURCE
Real assets	10.0%–20.0%	15.0%	Compliant	S3
Locked assets	At or below 10.0% of NAV	7.0%	Compliant	S3 S4
Private valuation age	At most 90 days	91 days	Near limit	S3 S4

03 Current Real Assets Portfolio

REAL-ASSETS ALLOCATION BY SUB-SEGMENT



CURRENT SLEEVE

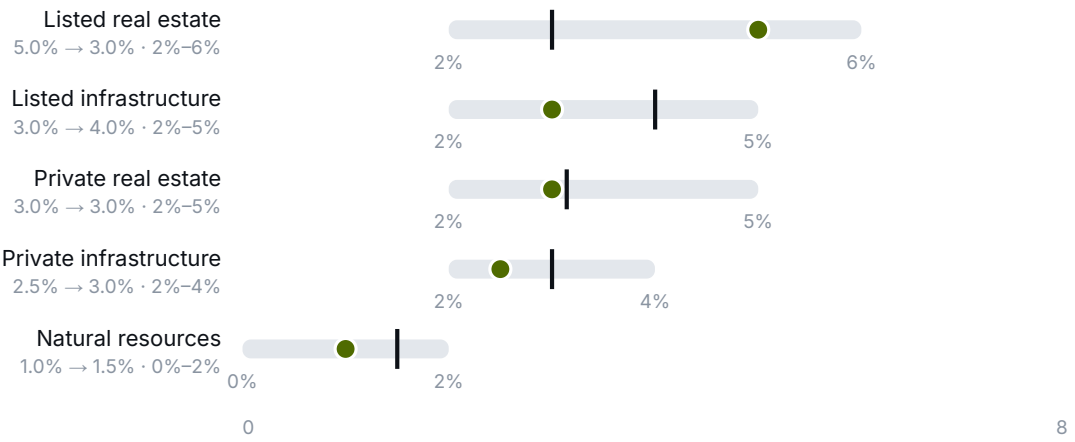
SUB-SEGMENT / EXPOSURE	STRUCTURE	VALUE	UNFUNDED	% PORTFOLIO	AS OF
Listed real estate	Marketable	$50.0M	n/a	5.0%	2026-06-30 S4
Listed infrastructure	Marketable	$30.0M	n/a	3.0%	2026-06-30 S4
Private real estate	Drawdown	$30.0M	$12.0M	3.0%	2026-03-31 S4
Private infrastructure	Drawdown	$25.0M	$10.0M	2.5%	2026-03-31 S4
Natural resources	Mixed	$10.0M	$6.0M	1.0%	2026-03-31 S4
Commodity securities	Marketable	$5.0M	n/a	0.5%	2026-06-30 S4

Marketable exposures use returned market value; drawdown exposures use NAV. Unfunded commitments are shown separately and are not added to NAV S4.

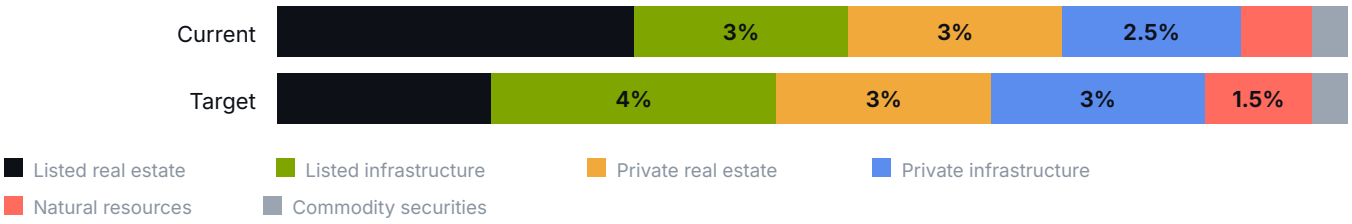
04 Target Sub-Segment Structure

Infrastructure and natural resources broaden the real-assets mix

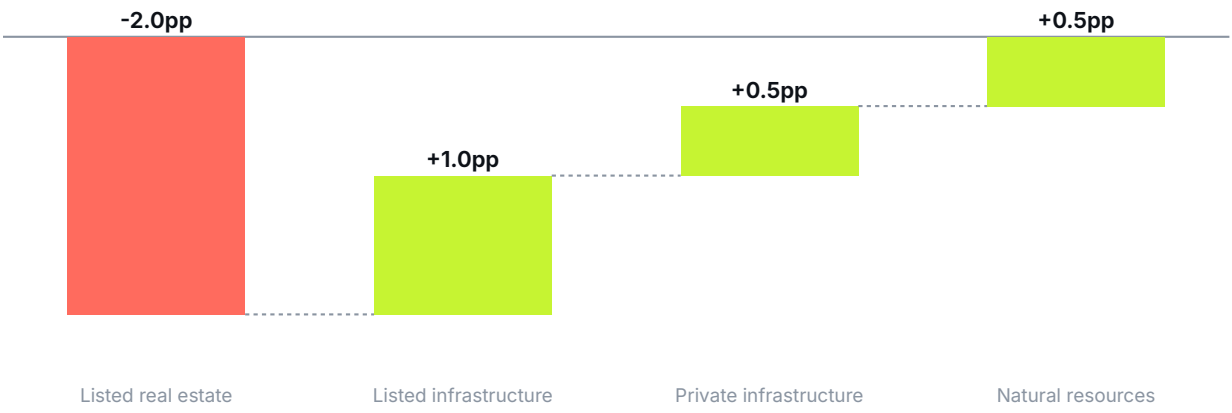
CURRENT WEIGHT VERSUS POLICY RANGE



CURRENT AND TARGET MIX



TARGET ALLOCATION CHANGES (PERCENTAGE POINTS)



CURRENT VERSUS TARGET

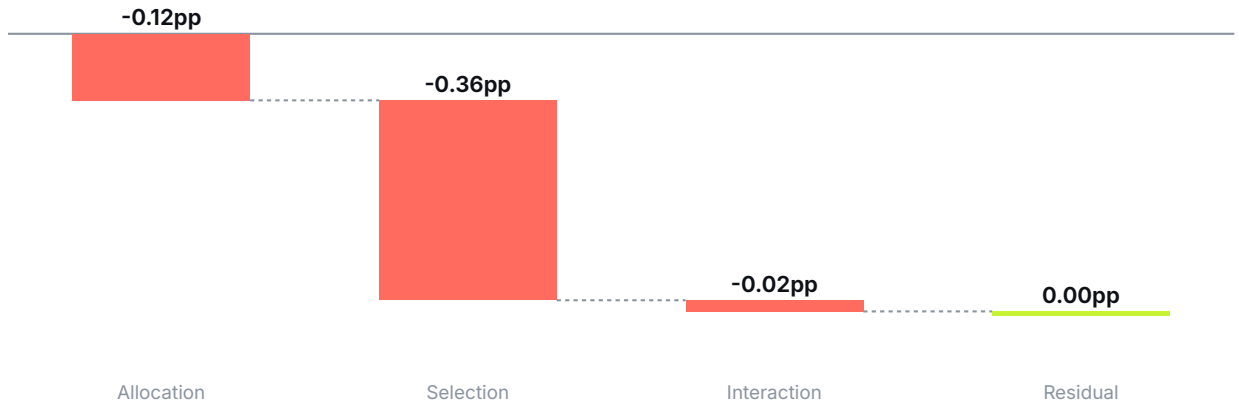
SUB-SEGMENT	CURRENT	TARGET	CHANGE	RANGE / CONSTRAINT	RATIONALE
Listed real estate	5.0%	3.0%	-2.0pp	2.0%–6.0%	Reduce largest sub-segment S3 S4
Listed infrastructure	3.0%	4.0%	+1.0pp	2.0%–5.0%	Increase marketable infrastructure role S3 S4
Private real estate	3.0%	3.0%	0.0pp	2.0%–5.0%	Maintain while valuations refresh S4 S8
Private infrastructure	2.5%	3.0%	+0.5pp	2.0%–4.0%	Phase after diligence S7 S8
Natural resources	1.0%	1.5%	+0.5pp	0.0%–2.0%	Broaden sub-segment mix S3 S7
Commodity securities	0.5%	0.5%	0.0pp	Mandate-specific	Maintain pending mapping review S8

> **Analysis — Why this target**
>
> Observation: The target shifts 2.0 percentage points from listed real estate toward infrastructure and natural resources S3 S4. Why it matters: The largest sub-segment declines under the proposed mix. Uncertainty: CMA, regime, findings, valuation and mapping coverage remain incomplete S8. What would change the view: Completed records showing no diversification improvement S4 S8.

The target is a committee proposal and is not a saved portfolio change S3 S8.

05 Historical Performance and Attribution

GOVERNED ATTRIBUTION EFFECTS



Returned attribution effects; residual is shown explicitly S5.

STANDARD PERIODS

PERIOD	PORTFOLIO / SLEEVE	BENCHMARK	EXCESS	COVERAGE
1Y real-assets sleeve	4.9%	5.4%	-0.5pp	Complete S5
3Y real-assets sleeve (ann.)	5.6%	5.2%	+0.4pp	Complete S5

ATTRIBUTION EFFECTS

EFFECT	CONTRIBUTION	SOURCE
Allocation	-0.12pp	S5
Selection	-0.36pp	S5
Interaction	-0.02pp	S5
Residual	0.00pp	S5

Governed attribution method and reconciliation metadata follow S5.

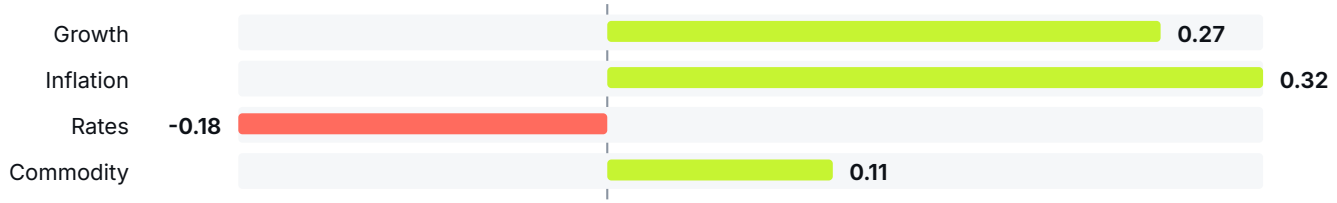
Method	**Brinson-Fachler** S5
Linking	**Symmetric Carino** S5
Basis	**Net, monthly linked** S5
Tolerance	**1 bp; reconciled** S5

06 Inflation, Commodity and Diversification Context

Positive sensitivities provide context, not guaranteed protection

HEADLINE INFLATION	INFLATION EXPOSURE	RATES EXPOSURE	INFLATION SENSITIVITY
2.8%	0.32	-0.18	Not available
2026-06-30 S7	Model-implied S7	Model-implied S7	No governed result S8

MODEL-IMPLIED REAL-ASSETS FACTORS



RETURNED CONTEXT

INDICATOR / EXPOSURE	OBSERVATION	CHANGE / SENSITIVITY	AS OF	SOURCE
Headline inflation	2.8%	-0.4pp year over year	2026-06-30	S7
Energy positioning	Net long index at 58th percentile	CFTC context only	2026-06-23	S7
Portfolio inflation sensitivity	Not available — no governed sensitivity result	Not assessed	2026-06-30	S8

Analysis — Context is not protection

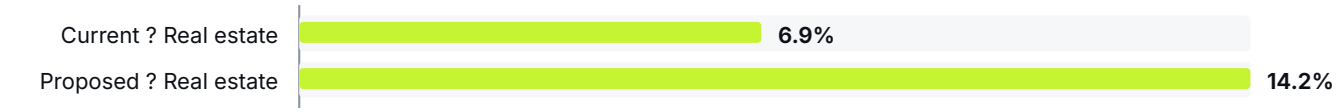
Observation: Two modeled exposures, inflation and commodities, are positive S7. Why it matters: These signals provide diversification context, not a guaranteed hedge. Uncertainty: Governed portfolio inflation sensitivity remains unavailable S8. What would change the view: Returned sensitivity results or mandate mappings contradicting both exposures S7 S8.

Macro and positioning evidence is context only; it does not establish inflation protection or explain portfolio performance S5 S7.

07 Forward Return and Risk

Governed assumptions frame tradeoffs rather than forecasts

FORWARD EXPECTED-RETURN COMPARISON



Governed forward assumptions; assumptions, not forecasts S6.

Assumption set	**2026 Strategic CMA** S6
Regime	**Baseline** S6
Horizon	**10 years** S6
Currency	**USD** S6
Context fingerprint	**test-ra-forward-2026** S6

REAL-ASSETS FORWARD ASSUMPTIONS

CLASS	EXPECTED RETURN	EXPECTED VOLATILITY
Real estate	6.9%	14.2%
Infrastructure	7.2%	11.8%
Natural resources	7.5%	16.8%

> **Analysis — Infrastructure balances risk**
> Observation: Infrastructure has lower expected volatility than real estate and natural resources in the selected assumptions S6. Why it matters: Its larger target role balances the broader mix. Uncertainty: Standalone governed mapping for commodity securities remains unavailable S8. What would change the view: Refreshed assumptions reversing the relative volatility results S6 S8.

FORWARD ASSUMPTIONS

SUB-SEGMENT	EXPECTED RETURN	EXPECTED VOLATILITY	MAPPING STATUS	SOURCE
Real estate	6.9%	14.2%	Governed class	S6
Infrastructure	7.2%	11.8%	Governed class	S6
Natural resources	7.5%	16.8%	Governed class	S6
Commodity securities	Not assessed — mapping not provided	Not assessed	No standalone CMA class	S8
Inflation-linked assets	Not assessed — mapping not provided	Not assessed	No standalone CMA class	S8

These figures are assumptions under the selected context, not forecasts or guarantees S6.

08 Commitments, Liquidity and Valuation

DRAWDOWN REAL-ASSETS CALLS AND DISTRIBUTIONS

PERIOD	CALLS	DISTRIBUTIONS
2026 H2	USD 9,000,000	USD 5,000,000
2027 H1	USD 8,000,000	USD 6,000,000
2027 H2	USD 6,000,000	USD 7,000,000

LIQUIDITY AND VALUATION

SUB-SEGMENT	STRUCTURE	LIQUIDITY / UNFUNDED	VALUATION DATE	STATUS
Listed real estate	Marketable	Daily	2026-06-30 S4	Available
Listed infrastructure	Marketable	Daily	2026-06-30 S4	Available
Private real estate	Drawdown	$12.0M unfunded	2026-03-31 S4	Near limit
Private infrastructure	Drawdown	$10.0M unfunded	2026-03-31 S4	Near limit
Natural resources	Mixed	$6.0M unfunded	2026-03-31 S4	Near limit

Liquidity and valuation coverage metadata follow S8.

Marketable share	**8.0% of NAV** S4
Locked share	**7.0% of NAV** S3
Unfunded ratio	**2.8% of NAV** S3
Private valuation lag	**91 days** S4
Base liquidity coverage	**6.2x** S3

09 Manager and Fund Diligence

MANAGER EVIDENCE

MANAGER / FUND	SUB-SEGMENT	EXPOSURE	DILIGENCE STATUS	OPEN FINDINGS	NEXT STEP
Example Real Estate Manager	Private real estate	$30.0M	Current S7	1 Medium S7	Close valuation follow-up
Example Infrastructure Manager	Private infrastructure	$25.0M	Current S7	No High findings S7	Refresh annual ODD
Proposed Natural Resources Manager	Natural resources	Not funded	Not assessed S7	Not assessed	Run comparison and ODD

Diligence condition

The proposed natural-resources manager requires comparison, operational diligence and performance-integrity review before funding S7.

10 Scenarios and Robustness

GOVERNED SCENARIO RESULTS



Inflation and denominator shocks are unavailable S8.

SUPPORTED SCENARIOS

SCENARIO	RETURN / LOSS	LIQUIDITY EFFECT	BASIS	SOURCE
Baseline assumptions	6.9% expected return	Base coverage 6.2x	Proposed sleeve	S3 S6
Inflation shock	Not available — no governed shock result	Not available	Unavailable	S8
Denominator stress	Not available — no governed stress result	Not available	Unavailable	S8

> **Selected-series sandbox — not saved-portfolio analytics**
> Not available — no Strategy Lab return series selected S8 .

11 Implementation Plan

IMPLEMENTATION STEPS

PHASE	ACTION	SIZE	FUNDING SOURCE	CONDITION	OWNER
1	Reduce listed real estate	2.0pp	Marketable sleeve	Transition review	Investment staff
2	Increase listed infrastructure	1.0pp	Sale proceeds	Policy recheck	CIO
3	Commit to private infrastructure	0.5pp	Sale proceeds and distributions	ODD and valuation refresh	Committee
4	Add natural resources	0.5pp	Sale proceeds	Manager approved	Committee

ALTERNATIVES CONSIDERED

ALTERNATIVE	BENEFIT	COST / RISK	REASON NOT SELECTED
No change	No transition	Listed real-estate concentration remains	Does not address structure objective S3 S4
Drawdown-only shift	Potential long-duration exposure	Higher lock-up and pacing risk	Marketable balance preferred S3 S4

12 Risks, Open Items and Approvals

DECISION RISKS

PRIORITY	RISK	EVIDENCE	MITIGATION / CONDITION	OWNER
High	Private valuation lag	91 days S4	Refresh valuations before commitment	Operations
Medium	Commodity and linker mapping	Not assessed S8	Document IPS taxonomy	CIO
Medium	Proposed manager diligence	Not assessed S7	Complete comparison and ODD	Investment staff

OPEN ITEMS AND APPROVALS

TIMING	ITEM	REQUIRED EVIDENCE	OWNER	STATUS
Before approval	Refresh private valuations	2026 Q2 NAVs	Operations	Open S4
Before approval	Confirm mandate mappings	IPS addendum	CIO	Open S8
Within 30 days	Select natural-resources manager	Comparison and ODD	Investment staff	Open S7
Next review	Refresh pacing	Returned commitment-pacing include	Private markets team	Open S6

13 Coverage

Portfolio and allocation	**Available**	Illustrative sample data	S2
Policy	**Available**	Allocation, liquidity and valuation-age rules	S3
Exposure	**Available**	Market value, NAV and unfunded	S4
Returns and attribution	**Incomplete**	Five-year sleeve period unavailable	S5
Forward assumptions and pacing	**Available**	Asset classes and drawdown charts	S6
Context and diligence	**Incomplete**	Proposed manager not assessed	S7
Mandate mappings and scenarios	**Not available**	Commodity, linker and shock mappings not returned	S8

A Appendix — Sources and methods

Based on 8 of 9 evidence areas.

TAG	EVIDENCE	SOURCE	AS OF	CHECK
S1	Organization and capabilities	list_organizations; get_gradient_capabilities	2026-06-30	**Passed**
S2	Portfolio and allocation	list_portfolios; get_portfolio_structure	2026-06-30	**Passed**
S3	Policy assessment	check_portfolio_policy	2026-06-30	**Passed**
S4	Real-assets exposure	get_portfolio_exposure	2026-06-30	**Passed**
S5	Returns and attribution	get_portfolio_historical_returns; get_portfolio_attribution	2026-06-30	**Incomplete**
S6	Forward assumptions and pacing	get_portfolio_exposure; get_capital_market_assumptions	2026-06-15	**Passed**
S7	Macro context and diligence	get_macro_conditions	2026-06-30	**Incomplete**
S8	Mandate mappings and gaps	get_gradient_capabilities	2026-06-30	**Advisory**

Illustrative, Gradient Maintained — demo data, not the client's holdings or managers. All organizations, portfolios, managers and results in this public example are illustrative.

SOURCE AUDIT TRAIL

TAG	KEY PARAMETERS	DIGEST
S1	organization=test	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S2	allocation_tree	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S3	portfolio=test	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S4	all pages	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S5	all sections	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S6	commitments; baseline	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S7	illustrative	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S8	illustrative	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa

Data scope for every source: Illustrative sample data. Each digest is the SHA-256 fingerprint of the exact data the report used.

B Appendix — Server Metric Methods and Disclosures

SERVER METRIC METHODS

SOURCE	QUANTITY	METHOD / FORMULA	VERSION	BASIS / UNITS / TOLERANCE
S3	Policy status	Governed assessment	portfolio-policy-v1	Returned status
S4	Exposure	Server exposure aggregation	portfolio-exposure-v1	Market value or NAV; unfunded separate

- Private valuations may lag the report date S4; forward values are assumptions, not forecasts S6.
- Drawdown charts cover closed-end sleeves only S6; commodity and linker mappings were not provided S8.
- Strategy Lab results are calculated separately. AI-assisted; not investment, legal or compliance advice.